

CALGARY INTL, AB, Canada

WMO: 718770

Lat: 51.114N

Lon: 114.020W

Elev: 1084

StdP: 88.96

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.5	-32.5	0.2	-27.6	-29.1	0.3	-24.3	14.2	1.0	12.8	-0.8	2.7	200	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.8	16.0	26.8	15.5	25.0	14.9	17.8	25.5	16.7	24.5	15.7	23.0	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	12.2	21.2	13.7	11.2	19.7	12.7	10.4	18.9	54.7	25.4	51.0	24.8	48.0	22.9	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.2	8.9	-30.8	32.0	3.5	1.7	-33.3	33.2	-35.3	34.2	-37.3	35.2	-39.8	36.4		
(5)			-31.0	20.0	3.4	1.7	-33.5	21.3	-35.5	22.3	-37.4	23.2	-39.9	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-6.9	-6.1	-2.0	4.3	9.8	13.7	17.1	16.2	11.6	5.2	-1.8	-6.5
	DBStd	10.59	8.93	7.88	7.91	5.20	4.62	3.23	3.02	3.34	4.60	5.10	7.10	7.94
	HDD10.0	2657	523	451	372	181	60	6	1	2	35	161	355	512
	HDD18.3	5054	781	684	630	422	265	143	60	82	204	408	604	771
	CDD10.0	689	0	0	1	8	54	117	220	193	84	12	1	0
	CDD18.3	43	0	0	0	0	1	4	21	15	2	0	0	0
	CDH23.3	774	0	0	0	2	36	59	309	277	89	2	0	0
	CDH26.7	177	0	0	0	0	3	9	78	71	17	0	0	0
(14) Wind	WSAvg	3.9	3.8	3.7	4.0	4.3	4.3	4.1	3.8	3.5	3.6	3.9	3.7	3.7
(15) Precipitation	PrecAvg	425	12	11	17	27	58	86	72	52	45	17	14	13
	PrecMax	629	28	34	59	79	144	177	159	167	148	48	44	37
	PrecMin	292	1	0	4	1	10	13	10	9	4	0	1	1
	PrecStd	69	8	8	11	18	29	42	35	34	32	11	10	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	12.2	16.6	22.2	26.7	28.0	31.2	31.3	29.1	22.2	16.2	12.2
		MCWB	4.6	4.6	6.8	10.1	12.8	16.0	17.5	16.0	14.2	10.9	7.4	3.6
	2%	DB	8.9	9.1	13.3	19.0	24.0	24.9	28.9	28.6	26.0	18.9	12.5	8.6
		MCWB	2.9	2.7	5.3	8.5	12.2	14.6	16.7	15.7	13.9	9.7	5.5	2.6
	5%	DB	7.1	6.6	10.9	16.0	21.3	22.8	26.8	26.5	23.2	16.0	10.0	6.4
		MCWB	1.8	1.3	4.0	6.9	11.0	13.7	16.7	15.4	12.9	8.0	3.8	1.4
	10%	DB	5.1	4.5	8.4	13.2	19.0	21.0	24.7	24.3	20.6	13.5	7.5	4.3
		MCWB	0.6	-0.1	2.7	5.4	10.0	12.7	15.9	14.9	11.7	6.6	2.4	-0.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	5.0	7.6	10.6	14.1	17.6	20.2	18.4	15.7	11.9	7.8	4.7
		MCDB	10.5	12.0	15.4	20.5	23.6	25.3	26.5	26.5	25.6	20.3	14.3	10.4
	2%	WB	3.2	3.0	5.8	9.0	12.7	15.8	18.6	17.2	14.2	10.0	5.9	2.9
		MCDB	8.3	8.6	12.7	18.1	22.2	22.5	25.7	25.5	23.9	17.8	11.9	8.4
	5%	WB	2.1	1.5	4.2	7.5	11.5	14.6	17.4	16.2	13.2	8.4	4.3	1.5
		MCDB	6.9	6.3	10.2	15.2	20.0	21.3	24.9	24.3	22.2	15.2	9.6	6.2
	10%	WB	0.7	0.1	2.7	5.9	10.3	13.5	16.3	15.2	12.1	6.9	2.6	0.0
		MCDB	5.0	4.2	8.1	12.2	17.9	19.6	23.5	23.0	20.1	12.9	7.2	4.1
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.5	10.6	11.4	11.8	11.0	12.3	12.8	12.4	11.3	10.0	10.1
		MCDBR	12.5	13.5	14.9	16.8	16.1	14.4	16.1	17.3	17.9	15.5	13.7	12.3
		MCWBR	8.3	8.7	8.7	8.9	7.8	6.9	7.1	7.4	8.3	8.4	8.2	8.1
	5% WB	MCDBR	12.3	13.2	14.3	15.8	14.6	13.1	14.2	14.9	16.6	15.0	13.6	12.4
		MCWBR	8.3	8.7	8.8	8.5	7.3	6.8	7.2	7.0	7.9	8.3	8.4	8.5
(40) Clear-Sky Solar Irradiance	taub		0.251	0.268	0.280	0.333	0.341	0.344	0.340	0.356	0.312	0.278	0.253	0.236
	taud		2.414	2.436	2.477	2.335	2.376	2.406	2.419	2.385	2.475	2.550	2.498	2.409
	Ebn at Noon		802	882	935	908	913	911	908	874	884	862	800	766
	Edh at Noon		57	74	87	113	114	112	108	107	87	66	53	49
(44) All-Sky Solar Radiation	RadAvg		1.13	2.11	3.39	4.66	5.51	5.89	6.47	5.29	3.73	2.29	1.23	0.88
	RadStd		0.08	0.16	0.25	0.34	0.42	0.42	0.39	0.37	0.40	0.24	0.09	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+73	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page